JAN–P–381

22 AUGUST 1946

JOINT ARMY-NAVY SPECIFICATION

POWDER, PROPELLENT, 4.2-INCH CHEMICAL MORTAR

Army Number
50-12-16C

Navy Number
4P13

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy supersedes the following specifications:

U. S. Army	Navy Department
50–12–16B 27 Mar. 1945	----------

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS, AND DRAWING.

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–156—Potassium Nitrate.
JAN–D–242—Diethylphthalate (For Use in Explosives).
JAN–N–244—Nitrocellulose (For Use in Explosives).
JAN–N–246—Nitroglycerin.
JAN–E–255—Ethyl Centralite (Carbamite).

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition, except Small Arms Ammunition.[1]
96–81–190—Charge, Propelling, 4.2 Inch, Chemical Mortar, M6.
100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATIONS

V–T–276—Thread, Cotton.
V–T–291—Thread, Linen.
V–T–301—Thread, Silk.
DDD–S–751—Stitches, Seams, and Stitching.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND PUBLICATION

Navy shipment Marking Handbook.[3]

INTERSTATE COMMERCE COMMISSION PUBLICATION

Regulations for Transportation of Explosives and other Dangerous Articles, etc.

A–3. *Drawing.*—The following drawing of the issue in effect on date of invitation for bids, forms a part of this specification:

U. S. ARMY ORDNANCE DEPARTMENT DRAWING

CP 24114—Heat Test.[4]

B. CLASSES.

B–1. This specification covers the following classes of propellent powder furnished in the form of approximately square, centrally perforated sheets as specified in the contract or order. (See par. H–1).

Class a. Half bundle.
Class b. Bundle.
Class c. Pack, five-bundle.

C. MATERIAL AND WORKMANSHIP.

C–1. *Materials.*—The raw materials used in the manufacture of 4.2-inch chemical mortar propellent powder purchased under this specification shall conform to the requirements of the applicable specifications listed in section A and be of the type, grade, class, or special composition as follows:

Nitrocellulose—Joint Army-Navy Specification JAN–N–244, grade C, type II.
Nitroglycerin—Joint Army-Navy Specification JAN–N–246 to which is added a maximum of 0.5 percent of methyl cellulose of a grade approved by the chief of the bureau or agency concerned.
Potassium nitrate—Joint Army-Navy Specification JAN–P–156, class c.
Thread, cotton—Federal Specification V–T–276, type IIIB, heavy silk (glazed) finish, unbleached thread, 6 ply, ticket number 10/6.
Thread, linen—Federal Specification V–T–291, type B, lock-stitch twist, 3-ply, left-hand twist thread.
Thread, silk—Federal Specification V–T–301, machine twist, commercial designation FF, white thread.

C–2. *Workmanship.*—

C–2a. *Process.*—Details of the manufacturing process and the equipment used by the contractor shall be submitted in writing to the bureau or agency concerned and written approval of the process received prior to the commencement of manufacture. Any deviation from the approved manufacturing process shall be submitted in writing and the written approval of bureau or agency concerned received prior to use.

[3] Applicable only to Navy purchases.
[4] Applicable only to Army purchases.

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C–2b. *Stitching.*—When linen, cotton, or silk thread is used, the stitching shall be done in accordance with Federal Specification DDD–S–751 for stitch type 301, 3 to 5 stitches per inch.

D. GENERAL REQUIREMENTS.

D–1. The powder covered by this specification shall be in the form of approximately square centrally perforated sheets adjusted to proper weight and otherwise ready for loading on the proper round without further treatment.

D–2. With every lot of powder submitted for acceptance, the contractor shall furnish, on official blanks, 8 copies of a descriptive sheet giving a complete history of its manufacture and results of chemical and physical analyses.

D–3. *Composition.*—The composition shall be as shown in table I.

TABLE I—*Composition*

Ingredient:	*Percent*
Nitrocellulose	52.15±1.50
Nitroglycerin	43.00±1.50
Diethylphthalate	3.00±0.50
Potassium nitrate	1.25±0.25
Ethyl centralite	0.60±0.20

D–4. *Volatiles.*—Maximum, 0.40 percent.

D–5. *Ash.*.—Maximum, 0.40 percent.

D–6. *Heat test requirement (120° C.).*—When tested as described in paragraph F–4d, complete fading of the methyl violet test paper to a salmon pink color shall not occur in less than 40 minutes. Fumes shall not be visible in 60 minutes test time.

D–7. *Sheet dimensions.*—The finished powder shall be granulated in the form of approximately square, centrally perforated sheets having the following average dimensions:

Approximate length of sides	2.7 inches.
Thickness	0.026±0.002 inch.
Diameter of central perforation	1.35±0.05 inches.

D–8. *Imperfect increments.*—There shall be not more than 1.0 percent of cracked, broken, torn, or otherwise imperfect increments.

D–9. *Ballistic requirements.*—

D–9a. When tested as specified in paragraph F–4g, the powder shall meet the following ballistic requirements:

Prescribed service velocity	820 ft/sec.
Rated maximum pressure (ball gage)	14,000 psi.
Maximum velocity variation	5 percent of prescribed.

D–9b. When the measured velocities and pressures for all rounds fired as specified in paragraph F–4g are plotted to scale as a function of the weight of the charge, the resultant curves shall be reasonably smooth.

E. DETAIL REQUIREMENTS.

E–1. *Individual increment weights.*—The increments shall be adjusted to the proper proportionate weight of charge established by the ballistic test (see par. F–4g) within the following tolerances:

Half bundles	1/51 charge ±1 grain.
Bundles	2/51 charge±2 grains.
Packs	10/51 charge±12 grains.

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Size of lots.*—Maximum, 20,000 pounds.

F–2. *Sampling.*—Each sample taken as specified in paragraphs F–2a to F–2c, inclusive, shall be packed in clean, dry airtight containers labeled to identify as to chemical, stability, physical or ballistic sample and to show the name of the material, lot number, date of sampling, name of manufacturer, contract number, plant and number of pounds in the lot. In the case of ballistic samples, each label shall show the number of the container from which the sample was taken. Chemical and ballistic samples shall be sent to the activity designated in the contract or order (see par. H–1). Stability samples shall be sent to the Commanding Officer, Picatinny Arsenal, Dover, N. J., or to the Supply Officer, Naval Powder Factory, Indian Head, Md.

F–2a. *Ballistic sample.*—Select a minimum of seven containers, or all containers if there are less than seven of each class of increment, in such a manner as to be representative of the lot. For each class of increment (half-bundle, bundle, pack) remove from the selected containers approximately an equal number of increments. The total number of each class of increment to be taken shall be determined by the number of each class required for a single propelling charge, times the number of rounds required in the ballistic test. These samples shall be packed in individual airtight containers.

F–2b. *Chemical and stability samples.*—Take from each container selected in accordance with paragraph F–2a, sufficient powder to form a representative sample of about 10 pounds.

F–2c. *Physical sample.*—Select a 5-pound sample so as to be representative of the various types of increments in the lot. This sample shall be used for physical measurements and weight determinations and shall be selected after the final packing.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made in accordance with U. S. Army Specification 50–0–1.

F–3b. *Navy.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—Tests shall be made in accordance with the following paragraphs and shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Composition.*—

F–4a(1). *Ether extractive matter.*—Cut into small pieces sufficient of the powder to make up 3 accurately weighed samples of approximately 5 gm. each. Transfer to Soxhlet extraction thimbles and extract with alcohol-free anhydrous ether. Use a hot water bath or a steam hot plate as the source of heat, and adjust the temperature so that the ether drips off the ends of the condensers at the rate of 2 to 3 drops per second. Continue the extraction until completion is indicated by the absence of residue when a portion of the ether is evaporated to dryness. When completion is established, evaporate the ether extracts by heating on a steam bath at low temperature until approximately 25 ml. of ether remains. Complete the evaporation to dryness by a slow current of dry air. Set aside the flasks for the determination of nitroglycerin as specified in paragraph F–4a(1)a, ethyl centralite as specified in paragraph F–4a(1)b, and diethylphthalate as specified in paragraph F–4a(1)c.

F–4a(1)a. *Nitroglycerin.*—

F–4a(1)a(1). *Preparation of solutions.*—Prepare the volumetric solutions that are necessary for this determination with freshly dis-

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tilled water which is free from atmospheric oxygen. Connect the siphon tubes and burets in such a way that only carbon dioxide gas, supplied from a Kipp generator, will be drawn into the stock bottles as the solutions are used. Replace the rubber tubing used for connections whenever longitudinal cracks appear. When no titrations are being made, turn off the stopcock in the tube connecting the Kipp generator with the stock bottle of titanous chloride solution and the buret. (See fig. 1.)

F–4a(1)a(2). *Titanous chloride solution.*—Prepare a 0.2 normal solution of titanous chloride by mixing 150 ml. of 20-percent titanium trichloride solution with 100 ml. of 38-percent hydrochloric acid solution for each liter of solution. The mixing operation should take place before diluting, and in these operations keep the solution protected from the air as much as possible by means of carbon dioxide. Mix the solution thoroughly with a current of carbon dioxide and store in a bottle covered with black paint or black paper to exclude light. To standardize the solution transfer on accurately weighed portion of 3.0 to 3.5 gm. of pure hexahydrated ferrous ammonium sulfate of known iron content to a 400-ml. beaker and dissolve in 50 to 75 ml. of water. Add 10 ml. of 38-percent hydrochloric acid solution and 0.5 gm. of potassium chlorate, then evaporate to dryness on a steam bath. Add 50 ml. of water, 5 ml. of 38-percent hydrochloric acid solution and evaporate again to dryness. Pass a current of carbon dioxide through a special type titration flask (see fig. 2) for 5 minutes to displace the air. Dissolve the resulting ferric salt in a minimum quantity of water and transfer to the flask. Add 25 ml. of 15-percent hydrochloric acid solution and titrate with 0.2 normal titanous chloride solution, keeping a current of carbon dioxide passing through the flask. Near the end point, which can be judged by the gradual disappearance of the reddish-colored ferric salt, add 5 ml. of 20-percent ammonium thiocyanate solution. Continue the titration until the blood-red color of ferric thiocyanate just disappears. Apply temperature and buret corrections to the observed reading.

$$\text{Normality of titanous chloride solution} = \frac{W}{0.3921\ V}$$

where

W = weight of ferrous ammonium sulfate.

V = corrected volume of titanous chloride solution.

Check the normality of the titanous chloride solution at least once a week by titration with ferric ammonium sulfate solution.

F–4a(1)a(3). *Ferric ammonium sulfate solution (0.15 N).*—Prepare a 0.15 normal ferric ammonium sulfate solution by mixing 75 gm. of hydrated ferric ammonium sulfate ($Fe_2(SO_4)_3.(NH_4)_2SO_4.24H_20$) with 25 ml. of 95-percent sulfuric acid for each liter of solution, and mix thoroughly by means of a current of carbon dioxide. Displace the air in a titration flask with carbon dioxide. Accurately measure 40 to 45 ml. of ferric ammonium sulfate solution into the flask, add 25 ml. of 15-percent hydrochloric acid solution and 50 ml. of water. Titrate with 0.2 normal titanous chloride solution until near the end point, then add 5 ml. of 20-percent ammonium thiocyanate solution. Continue the titration until the red color just disappears. Apply temperature and buret corrections to the observed readings and calculate the normality of the ferric ammonium sulfate solution. It can be assumed that the strength of the solution will remain constant for an indefinite period.

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F–4a(1)a(4). *Ferrous chloride solution (0.7 N).*—Prepare a 0.7 normal ferrous chloride solution by mixing 140 gm. of ferrous chloride tetrahydrate with 50 ml. of 38-percent hydrochloric acid solution for each liter of solution, and mix thoroughly by means of a current of carbon dioxide. This solution invariably contains a small amount of ferric iron, which must be determined by running a blank under the same conditions as in the analysis for nitroglycerin. Displace the air from a titration flask with a current of carbon dioxide. Add an accurately measured portion of 15 ml. of 0.7 normal ferrous chloride solution, 25 ml. of 15-percent hydrochloric acid solution, 25 ml. of glacial acetic acid, and boil for 5 minutes under a reflux condenser on a hot plate. Cool to room temperature by means of a large beaker of cold water, increasing the current of carbon dioxide so that no air will be sucked down into the flask. It is convenient to hold the index finger over the top of the condenser until equilibrium is established. Disconnect the condenser and add 5 ml. of 20-percent ammonium thiocyanate solution, then titrate with 0.2 normal titanous chloride solution until the red color just disappears. This volume of titanous chloride solution, usually 0.5 to 1.0 ml. must be subtracted from subsequent titrations involved in the determination of nitroglycerin. Run a blank at frequent intervals to determine whether the ferrous chloride solution has changed. Whenever any new bottles of reagents are used, run a new blank to detect any interfering impurities in these reagents.

F–4a(1)a(5). *Procedure.*—Dissolve one of the dried residues (see par. F–4a (1)) with glacial acetic acid, transfer the solution to a 250-ml. volumetric flask, make up to volume with glacial acetic acid and mix thoroughly. Avoid temperature changes on account of the high thermal coefficient of expansion of acetic acid. Displace the air in a titration flask by passing in a current of carbon dioxide for 5 minutes. Rinse a calibrated 25-ml. pipet with the acetic acid solution of the sample, then transfer a 25 ml. portion of the solution to the flask. Add an accurately measured portion of 15 ml. of 0.7 normal ferrous chloride solution and 25 ml. of 15-percent hydrochloric acid solution, connect the flask to a reflux condenser, and boil the solution gently for 5 minutes on a hot plate. Add a few glass beads to prevent possible bumping. The color of the solution will change from yellow to dark green to reddish-brown as the reaction proceeds. Increase the current of carbon dioxide, then cautiously immerse the flask in a large beaker of cold water, keeping the index finger over the top of the condenser until the hot vapors are condensed. After cooling the flask and contents to room temperature, disconnect the condenser, insert a one-hole rubber stopper which contains a short piece of glass tubing. Insert the tip of the buret into the glass tubing, and titrate with 0.2 normal titanous chloride solution until near the end point, as shown by disappearance of the reddish color of ferric iron. Add 5 ml. of 20-percent ammonium thiocyanate solution and continue the titration until the deep color of ferric thiocyanate just disappears. Apply temperature and buret corrections to the observed reading. Calculate the nitroglycerin content on a volatiles-free basis.

$$\text{Percent nitroglycerin} = \frac{2.523(V-B)(N)}{W}$$

where

V = volume of titanous chloride solution used.
N = normality of titanous chloride solution.

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B=volume of titanous chloride solution used for titration of blank made on reagents as directed above in the standardization of ferrous chloride solution.

W=weight of sample in aliquot corrected for volatiles.

The titrations on duplicate aliquot portions should check with in 0.05 ml. In the above calculation, the molecular weight of nitroglycerin is divided by 9 to obtain the equivalent weight.

F–4a(1)b. *Ethyl centralite.*—

F–4a(1)b(1). *Reagents and apparatus.*—

F–4a(1)b(1)a. *0.2 N $KBrO_3$ solution containing 30 gm. KBr per liter.*—Dry a quantity of chemically pure $KBrO_3$ for 2 hours at 100° C. Weigh out 5.567 gm. Dissolve in distilled water along with 30 gm. of KBr and make up the solution to a liter in a volumetric flask. The solution will keep indefinitely.

F–4a(1)b(1)b. *1:1 HCl.*—Dilute 35–37-percent HCl with an equal amount of water.

F–4a(1)b(1)c. *10-percent KI solution.*—Dissolve 100 gm. of KI in 900 gm. of distilled water.

F–4a(1)b(1)d. *Carbon tetrachloride.*—Run a blank determination on each lot of carbon tetrachloride in order to correct for errors from this source.

F–4a(1)b(1)e. *Flasks.*—Flasks shall be stoppered iodine titration type of 250 ml. capacity, equipped with a gutter.

F–4a(1)b(1)f. *Starch solution.*—To each liter of cold distilled water, add 5 gm. of cornstarch made into a paste with a little water. Add 10 gm. of KI and 4 gm. of $ZnCl_2$ crystals. Bring the mixture to boiling with agitation. Simmer for 10 minutes and continue agitating. Cool and transfer to a large bottle provided with a siphon. The starch will settle to the bottom. Use only the supernatant liquid for the determinations. The end point is sharp and the depth of the blue starch-iodide color increases with age to a very intense blue.

F–4a(1)b(1)g. *0.11N $Na_2S_2O_3$.*—Dissolve approximately 30 gm. of $Na_2S_2O_3.5H_2O$ in a liter of distilled water. The solution should be allowed to stand for a few weeks before being standardized against the 0.2 N $KBrO_3$-KBr solution. The standardization shall be conducted as follows: Add 50 ml of carbon tetrachloride to the iodine flask and add 25 ml of $KBrO_3$-KBr solution from a buret or accurate pipet. Moisten the stopper of the flask with a drop of KI solution. Add 10 ml. of 1:1 HCl to the flask, stopper quickly, and shake for 5 minutes. Measure out 25 ml. of 10-percent KI solution. Cool the flask under the cold water tap. Add approximately 5 ml. of KI solution to the gutter of the flask, then loosen the stopper and allow the solution to run into the flask, taking as much care as possible to prevent the loss of bromine fumes. Add the remainder of the KI solution, stopper, and shake the flask for 30 seconds. Titrate the liberated iodine with the $Na_2S_2O_3$ solution. The solution finally changes from a brown to a deep violet color which on further addition of $Na_2S_2O_3$ becomes less intense. Near the end point, add a few ml of starch solution and continue the titration with frequent shaking until the blue color is completely discharged. Calculate the normality of the $Na_2S_2O_3$ solution as follows:

$$\text{Normality}=\frac{25 \times 0.2}{V}$$

where V=ml. of $Na_2S_2O_3$ solution used in titration.

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F–4a(1)b(2). *Procedure.*—Dissolve one of the dried residues (see par. F–4a(1)) in 10 to 15 ml of carbon tetrachloride and transfer to a 100 ml. volumetric flask. Rinse with several 10 ml. portions of carbon tetrachloride and add to the contents of the volumetric flask. Make up to volume and pipet a 25 ml. aliquot portion into the iodine titration flask. Add an additional 25 ml. of carbon tetrachloride. Add 25 ml. of the $KBrO_3$-KBr solution measured from a buret or an accurate pipet. Moisten the stopper with a drop of KI solution, add 10 ml. of 1:1 HCl and quickly stopper the flask, then shake thoroughly for 1 minute. Immediately at the end of this period, add the KI and titrate the liberated iodine, following the procedure outlined in paragraph F–4a(1)b(1)g. Do not allow the mixture to stand after the initial shaking period nor after the addition of the KI, but always complete the determination without delay. Calculate the percent of ethyl centralite as follows:

$$\text{Percent ethyl centralite} = \frac{6.70(V-v)N}{W}$$

where

V=ml $Na_2S_2O_3$ used in blank (see par. F–4a(1)b(1)g).
v=ml $Na_2S_2O_3$ used in titration of sample.
N=normality of $Na_2S_2O_3$ (see par. F–4a(1)b(1)g).
W=weight of volatiles-free sample in aliquot.

To insure complete bromination of the ethyl centralite at least a 100-percent excess of $KBrO_3$-KBr should be present. That is "v" should be equal to or greater than "$(V-v)$".

F–4a(1)c. *Diethylphthalate.*—

F–4a(1)c(1). *Apparatus.*—Prepare a chromatographic column as shown in figure 3 or by fitting a piece of pyrex (or equal) thick-walled glass tubing 28 mm. inside diameter by 5 inches long into a 30-ml. size fritted glass funnel containing a small cushion of glass wool. The connection between the tube and the funnel is made by means of a piece of Gooch type tubing. Fit the funnel into a 500-ml. side arm suction flask which is connected to a water aspirator. With the aspirator on full force, add a 2:1 silicic acid-Celite (Merck or equal) adsorbent to the column by means of a spatula. When the column is full, firmly push down the adsorbent with the aid of a cork which barely fits into the tube. The top of the column is then fitted with a cork bearing a 200-ml. separatory funnel and a piece of capillary tubing extending from a point above half the height of the separatory funnel to reach ¼ inch into the tube. The cork shall make an airtight seal with the column.

F–4a(1)c(2). *Adsorption.*—Dissolve one of the dried residues from the ether extract determination in sufficient 1:1 benzene-ligroin solution to make a volume of 40 ml. (dissolve the sample in benzene and then add ligroin). Pass, in order, the following solutions into the column while the latter is under full suction of the water aspirator: (The solutions shall not touch the cork nor shall the solution be allowed to be sucked below the top of the adsorbent between additions.)

85 ml. anhydrous ether
85 ml. ligroin
40 ml. benzene-ligroin solution of the sample prepared above
5 rinsings of the sample flask with 5 ml portions of benzene-ligroin
175 ml. benzene-ligroin
85 ml. ligroin

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Maintain the vacuum for 5 minutes after the final ligroin has passed into the top of the column.

F–4a(1)c(3). *Solution, precipitation and calculation.*—Remove the adsorbent with a wooden rod which just fits into the tube. Discard the thin, colored layer at the very top of the column (about 1 mm. thick) and transfer about 3¼ inches of the column, immediately below the colored layer, to a 100-ml. fritted glass filtering funnel fitted to a bell jar filtration apparatus. Add 40 ml. of dry ether to the adsorbent in the funnel. Mix the adsorbent with the ether by stirring with a glass rod. Remove the ether by applying suction and catch the filtrate in a 250-ml. beaker. Repeat the extraction with three more 40-ml. portions of dry ether. Evaporate the ether at about 40° C. to about 50 ml. with the aid of a gentle current of dry air, then transfer the ether to a ground glass stoppered 125 ml. Erlenmeyer flask. Evaporate the ether and dry the residue for 15 minutes in a vacuum desiccator. Make sure that no moisture remains in the flask. Pipet into the flask, 10 ml. of a previously filtered 0.5 *N* solution of potassium hydroxide in absolute alcohol. Loosely stopper the flask and heat for 1 hour on an open steam bath. After the contents of the flask have been heated to the temperature of the bath, release the pressure in the flask and tightly close the glass stopper. After 1 hour, remove the flask from the bath, loosen the stopper and cool in a water bath, release the pressure in the flask and tightly close the glass stopper. After 1 hour, remove the flask from the bath, loosen the stopper and cool in a water bath to room temperature. Transfer the precipitate of dipotassium phthalate onto a tared asbestos mat crucible and wash with 10-20 ml. of absolute alcohol. Particles in the flask may be transferred with the use of a rubber policeman. Dry the crucible at 150° C. for 2 hours and weigh. Calculate the percent diethylphthalate as follows:

$$\text{Percent diethylphthalate} = \frac{91.7\ (A - 0.0100\ \text{gm})}{W}$$

where

A = gm of precipitate
W = gm of sample corrected for volatiles.

F–4a(1)c(4). *Correction factor.*—The correction factor of 0.0100 gm., applied in the calculation of the percent of diethylphthalate, in paragraph F–4a(1)c(3), is the amount of potassium nitrate derived from the organic nitrate impurities present in the nitroglycerin by decomposition with alcoholic potassium hydroxide. This factor should be checked frequently as follows: Dissolve the precipitate from the filtering crucible by sucking five 10 ml. portions of water through the crucible into a 150-ml. side arm suction flask. Quantitatively transfer the filtrate to a 100-ml. volumetric flask and dilute to the mark. Transfer a 5 ml. aliquot to a 50-ml. beaker by means of a pipette. Evaporate the 5 ml. to dryness on a steam bath. Cool the beaker, and add 2 ml. of phenoldisulfonic acid reagent. (See par. F–4a(1)c(5).) Mix thoroughly with a stirring rod to insure intimate contact, and after 5 minutes, dilute with 25 ml. of water. Neutralize the solution with 1:1 ammonium hydroxide and continue to add ammonium hydroxide until the maximum color has been developed. Transfer the solution to a 50-ml. volumetric flask and dilute to the mark. Determine the amount of potassium nitrate present colorimetrically by comparison with a standard potassium nitrate solution containing 0.100 gm. of potassium

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nitrate in 1 liter of water. A 5 ml. aliquot of the standard solution is treated in a similar manner as the unknown solution.

F–4a(1)c(5). *Preparation of phenoldisulfonic acid reagent.*—Dissolve 6 gm. of phenol in 37 ml. of H_2SO_4. Heat on a steam bath until solution is complete. Cool and add 3 ml. of water.

F–4a(2). *Potassium nitrate.*—

F–4a(2)a. *Preparation of reagent.*—Dissolve 200 gm. of chemically pure trisodium cobaltinitrite $Na_3Co(NO_2)_6.H_2O$ in water and dilute the solution to 500 ml. Store the solution in a dark bottle and avoid direct sunlight. This reagent is stable for only one week. Reagents older than one week should be discarded, as they tend to give high results.

F–4a(2)b. *Procedure.*—Weigh a 2 gm. sample into a tared 90 mm. evaporating dish, add 10 ml. of 70 percent HNO_3 and place on the steam bath until the reaction starts. Turn off the steam until the reaction subsides, then evaporate to a gummy mass. (Do not take to dryness on a hot plate.) Place the dish on a nichrome triangle on a tripod, and heat cautiously from the top by means of a burner. Finally heat from the bottom until the bulk of the carbonaceous matter is removed, then transfer to a muffle furnace at dull red heat for 10 to 15 minutes. A small amount of carbon can be neglected. Allow the dish to cool, rinse the sides with 5 ml. of water, then add 5 ml. of concentrated HCl. Cover the dish with a watch glass, place the dish on the steam bath for 5 to 10 minutes, dilute the contents of the dish to approximately 25 ml., filter the solutions through a filter paper into a 150 ml. beaker, and wash the residues thoroughly. Save the residue for the ash determination. (See par. F–4b.) Evaporate the filtrate to dryness on the steam bath. Add 25 ml. of 1 *N* HNO_3, and cool to 20° C. By means of a graduate, add 15 ml. of cooled (20° C.) 2 *N* trisodium cobaltinitrite solution (see par. F–4a(2)a) with stirring, to the cooled solution containing potassium ions. Cover the resulting mixture with a watch glass and allow to stand for 2 hours at 20° C. Filter the solution through a tared filtering crucible. Use a minimum amount of 0.01 normal nitric acid solution in a wash bottle to transfer the precipitate of dipotassium-sodium cobaltinitrite from the beaker to the crucible. Wash the precipitate with three 10 ml. portions of 0.01 normal nitric acid solution and then with three 5 ml. portions of alcohol. Aspirate the crucible until the contents are fairly dry, place the crucible and contents in an oven at 100° C. for 1 hour, cool in a desiccator and weigh.

$$\text{Percent potassium nitrate} = \frac{44.5A}{W}$$

where

A = weight of precipitate.
W = weight of the volatiles free sample.

F–4a(3). *Nitrocellulose.*—Calculate the percentage of nitrocellulose by subtracting from 100 percent the sum of the percentages of all other ingredients.

F–4b. *Ash.*—Transfer the filter paper used in making the potassium nitrate determination in accordance with paragraph F–4a(2)b, to a previously ignited and weighed crucible, dry for 30 minutes at 100° C., char the paper by means of a burner, ignite in a muffle furnace at dull red heat for 15 to 30 minutes, cool in a desiccator and weigh. Calculate the increase in weight as percent ash in the powder.

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F–4c. *Volatiles.*—Weigh accurately a 10 gm. portion of the sample cut into small pieces and place in a tared aluminum dish with a glass cone-shaped cover. (See fig. 4.) Place the apparatus on a closed steam bath and heat for 2 hours. Cool the apparatus in a desiccator containing calcium chloride and weigh. Calculate the loss in weight as percent volatiles.

F–4d. *Heat test (120° C.).*—Weigh out 5 portions of the sample, each weighing 2.5 gm., and place each in a heavy glass test tube (Pyrex or equal), approximately 15 mm. inside diameter, 18 mm. outside diameter and 290 mm. long. Place a piece of standard normal methyl violet paper,[5] 70 mm. long and 20 mm. wide, vertically in each tube, its lower edge 25 mm. above the powder. Stopper the tubes with corks through which holes 4 mm. in diameter have been bored. Place the tubes in a constant temperature bath of 120° ± 0.5° C. (see Army Ordnance Department Drawing CP–24114 for apparatus used at Army activity) so that no more than 6 or 7 mm. of length projects. Examine each tube by withdrawing one-half of its length and replacing quickly at 5-minute intervals after 35 minutes have elapsed. Record as the time of completion of the test, the time of the observation which reveals the test paper in any tube to be completely changed to a salmon pink color. At the end of 60 minutes of heating, remove the tube from the bath, place in front of a white paper, and note if any fumes are present in the tube.

F–4e. *Sheet dimensions.*—

F–4e(1). *Length of side.*—Select at random 30 increments of powder, taking the number in approximate proportion to the weight of the different type increments in the lot. Only one sheet from any increment shall be used for measurement. Measure the length of each side of each sheet by means of a graduated steel scale. Calculate the average length of side.

F–4e(2). *Diameter of central perforation.*—Using the same sheets selected and used as specified in paragraph F–4e(1), measure the diameter of the central perforation in each along two axes at right angles to each other. Use a measuring microscope having a magnification of 25 diameters equipped with cross hairs, and having a horizontal scale and vernier screw scale capable of giving measurements accurate at least to 0.01 inch. Calculate the average diameter of the central perforation.

F–4e(3). *Thickness.*—Using the same proportions of increments as selected in paragraph F–4e(1), determine the thickness of each of 60 sheets of powder by means of a micrometer. Calculate the average thickness.

F–4f. *Imperfect increments.*—Weigh the physical sample and separate all cracked, broken, torn, or otherwise imperfect increments. Weigh these and calculate the percentage of imperfect increments.

F–4g. *Ballistic tests.*—

F–4g(1). Establish a charge of the powder which when fired in a 4.2-inch chemical mortar M2 with a chemical shell M2 loaded to a weight of 25.50 pounds using the M2 fuze and Mk. I type I, class A, ignition cartridge, will give the prescribed service velocity (see par. D–9a) with a mean maximum pressure not exceeding the rated maximum pressure. Each of the charges fired shall be composed of increments so selected as to be as nearly representative of the lot as possible.

[5] Secure standard normal methyl violet paper from the Office, Chief of Ordnance, War Department, Washington 25, D. C., or Naval Powder Factory, Indian Head, Md.

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F–4g(2). *Velocity and pressure uniformity.*—Fire seven rounds using powder from a different sample container for each of these rounds, i. e., packs to be taken from an individual sample container and bundles from an individual sample container. The charge will be made up in the same manner as a service charge as specified in U. S. Army Specification 96–81–190 using 4 packs, 5 bundles, and 1 half-bundle.

F–4g(3). *Graph.*—Plot to scale, the measured velocities and pressures for all rounds fired as a function of the weight of the charge. Examine the resulting graph to ascertain whether the points lie on reasonably smooth curves.

F–4g(4). In case the results indicate that the nonuniformity of either velocity or pressure in one round is due to defects other than those existing in the powder, an additional round may be fired, the results of which will replace in the uniformity group the values obtained in the round which gave the inconsistent results. The values obtained with the round which was not included in the uniformity group shall, however, be shown on the firing records.

F–5. *Retest.*—Should a lot fail to pass the specified tests (either ballistic, chemical, or physical) it shall be rejected, but it may be retested once, upon application from the manufacturer and approval of the bureau or agency concerned.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packing.*—The finished powder shall be packed in containers complying with Interstate Commerce Commission Specification 14 and holding not more than 75 pounds of powder. The half bundles, bundles and packs shall be in separate containers. The approximate proportion of each as it exists in the service round will be maintained in the final lot.

G–2. *Marking.*—

G–2a. *Army.*—Unless otherwise specified in the contract each container shall be marked in accordance with U. S. Army Specification 100–2 with additional markings giving place and date of completion of manufacture, name of manufacturer, class of powder, lot number, box number, and the destination of the weapon for which the powder is intended.

G–2b. *Navy.*—In addition to any special markings required by the contract or order, shipments for the Navy shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules and contracts or orders should specify the following:

(a) The title, number and date of the specification.
(b) The class required. (See par. B–1.)
(c) The activity to which samples should be sent. (See par. F–2.)

H–2. *Use.* The powder covered by this specification is intended for use in propelling charges for the 4.2-inch chemical mortar.

H–3. Copies of Army Ordnance Department drawings may be obtained upon application to Office, Chief of Ordnance, War Department, Washington 25, D. C.

H–4. Copies of Interstate Commerce Commission Regulations for Transportation of Explosives and other Dangerous Articles, of which I. C. C. Specification 14 forms a part, may be obtained upon application accompanied by postal note, money order, coupon, or cash to the

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Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 40 cents.

H–5. Copies of Joint Army-Navy Specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases) and Navy Department specifications and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office of the Chief of Ordnance, War Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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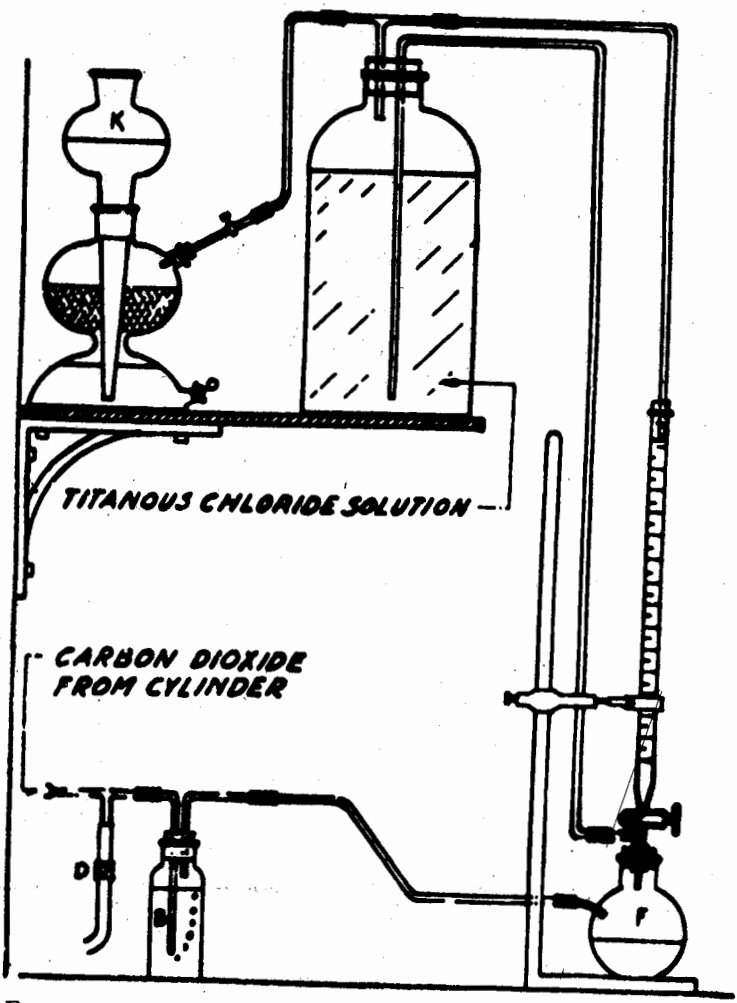


FIGURE 1.—Apparatus for storing and using titanous chloride solution.

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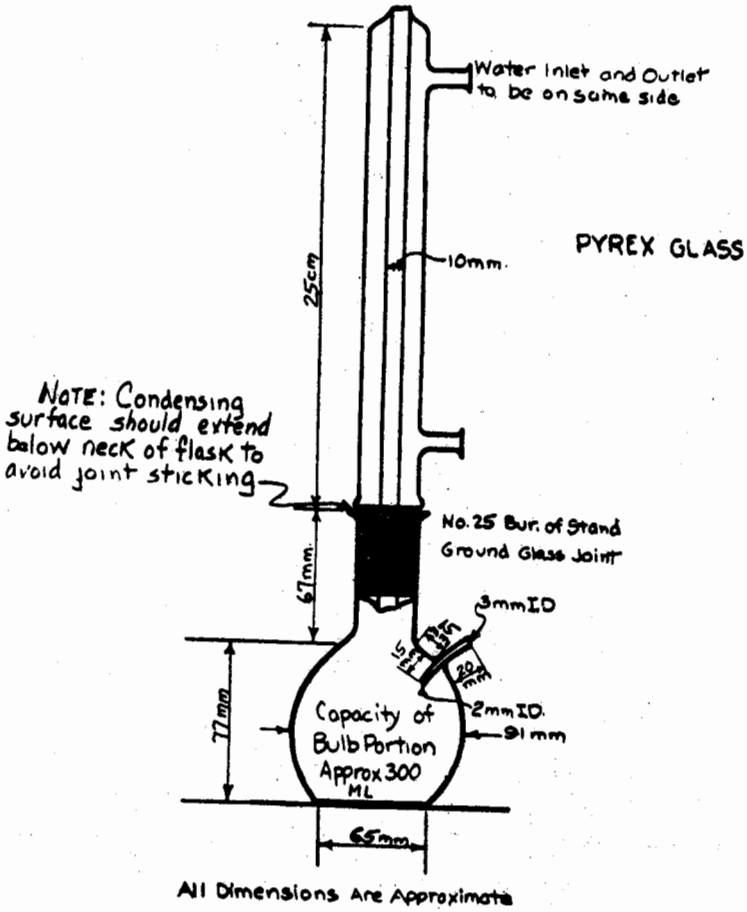


FIGURE 2.—Titration flask.

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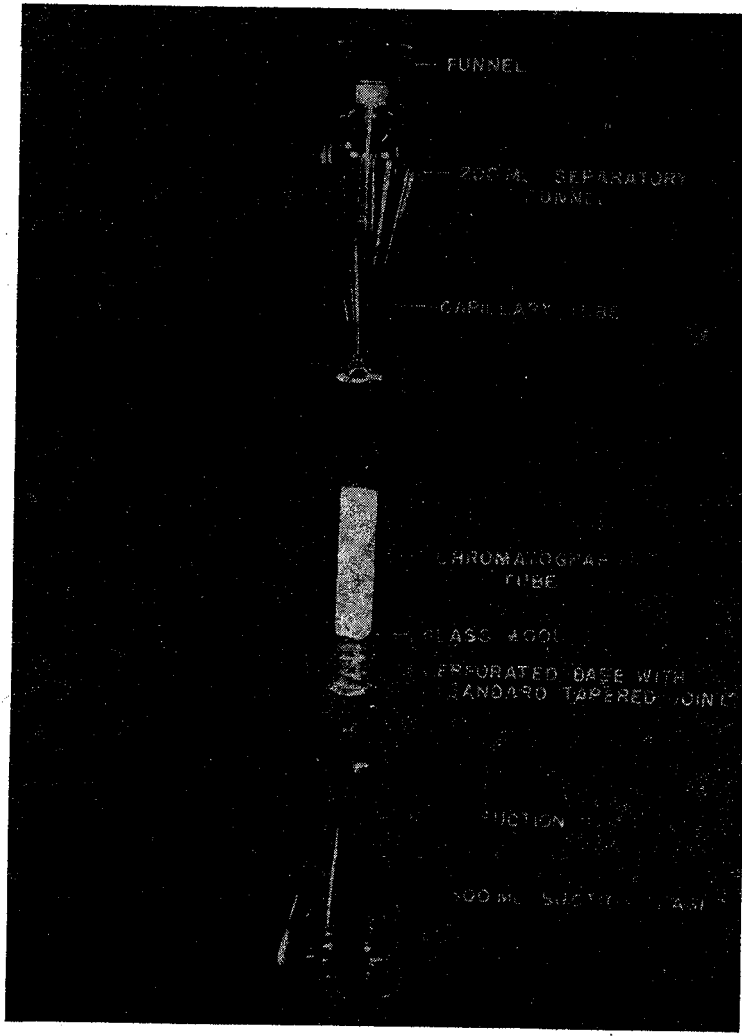


FIGURE 3.—Chromatographic column.

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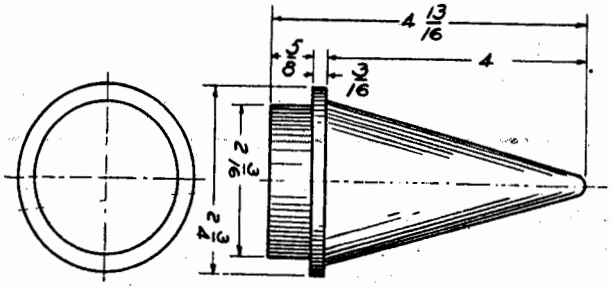


FIGURE 4.—Aluminum dish with cone-shaped cover.

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